**THE UNIVERSITY OF HONG KONG**
**DEPARTMENT OF MATHEMATICS**
**MATH3603 Probability theory**

Tutorial 2

1. Bill and George go target shooting together. Both shoot at a target at the same time. Suppose Bill hits the target with probability 0.6, where as George , independently, hits the target with probability 0.3.

   (a) Given that exactly one shot hit the target, what is the probability that it's George's shot?

   (b) Given that the target is hit,what is the probability that George hit it?

2. Three prisoners A, B and C are informed by their jailer that one of them has been chosen at random to be executed, and the other two are to be freed. Prisoner A asks the jailer to tell him privately which of his fellow prisoners will be set free, claiming that there would be no harm in divulging this information, since he already knows that at least one will go free. The jailer refuses to answer this question, pointing out that if A knew which of his fellows were to be set free, then his own probability of being executed would rise from 1/3 to 1/2 since he would then be one of the prisoners. Is the jailer correct?

3. A bowl contains twenty cherries, exactly fifteen of which have had their stones removed. A greedy pig eats five whole cherries, picked randomly, without any remarks on the presence or absence of the stones. Subsequently, a cherry is picked randomly from the remaining fifteen.

   (a) What is the probability that this cherry contains a stone?

   (b) Given that this cherry contains a stone, what is the probability that the pig consumed at least one stone?

4. The $n$ passengers for a Bell-Air flight in an airplane with $n$ seats have been told their seat numbers. They get on the plane one by one. The first person sits in the wrong seat. Subsequent passengers sit in their assigned seats whenever they find them available. or otherwise in a randomly chosen empty seat. What is the probability that the last passenger finds its seat free.